UNIT V

Transport Level Security: Web Security Requirements, Transport Layer Security (TLS), HTTPS, Secure Shell(SSH)

Firewalls: Firewall Characteristics and Access Policy, Types of Firewalls, Firewall Location and Configurations.

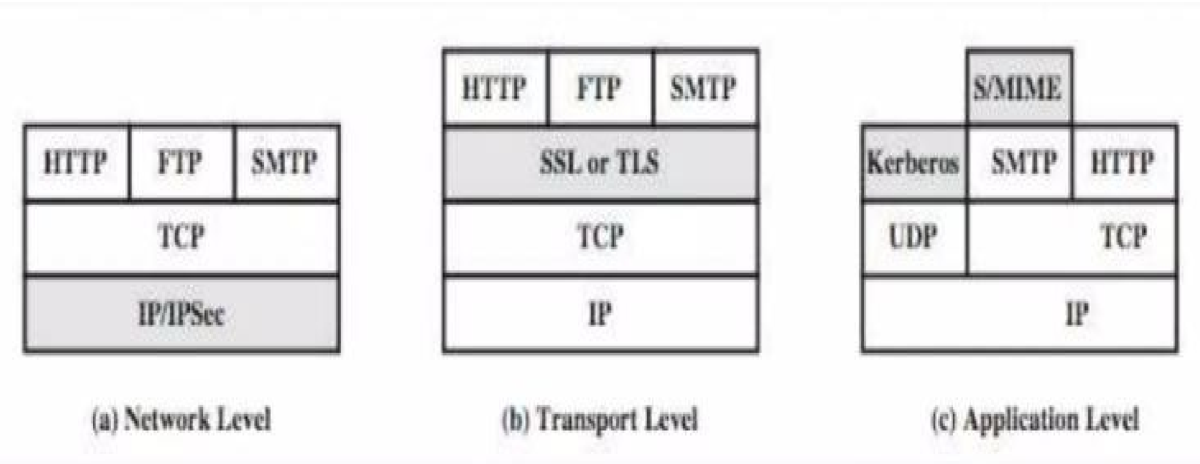
Web Security:

- Web security refers to the protection of data as it travels across the internet or within a network.
- It plays vital role in websites, web applications, and the servers they run on from malicious attacks.
- widely used by business, government, individuals.
- To secure your data in web security need to consider the following things:
    - Keep Software Updated
    - Validate User Input
    - Use Strong Passwords
    - Enable Two-Factor Authentication
    - Monitor and Log Activity

Web Security Requirements:

- Requirements of Web Security:
- 1.Confidentiality: Ensuring that data remains private and is only accessible by authorized parties.
- 2.Integrity:Guaranteeing that data is not tampered with or altered during transmission.
- 3. Authentication: Verifying the identities of communicating parties to ensure trustworthiness.
- 4. Non-repudiation: Ensuring that a sender cannot deny the transmission of a message.
- 5.Availability: Making sure that resources and services are accessible when needed.

Web Security Approaches:



Transport Level Security:

- TLS stands for Transport Layer Security.
- (TLS) is a protocol that ensures privacy between communicating applications and their users on the Internet.
- When a server and client communicate, TLS ensures that no third party may eavesdrop or tamper with any message.
- TLS is the successor to the Secure Sockets Layer (SSL).
- TLS is composed of two layers:
- TLS Record Protocol.
- TLS Handshake Protocol.
- TLS Record Protocol provides connection security Using
- Symmetric key algorithms.
- The TLS Handshake Protocol allows the server and client to authenticate each other and also share the details about encryption algorithm and cryptographic keys before data is exchanged.

Transport Level Security or Secure Socket Layer:

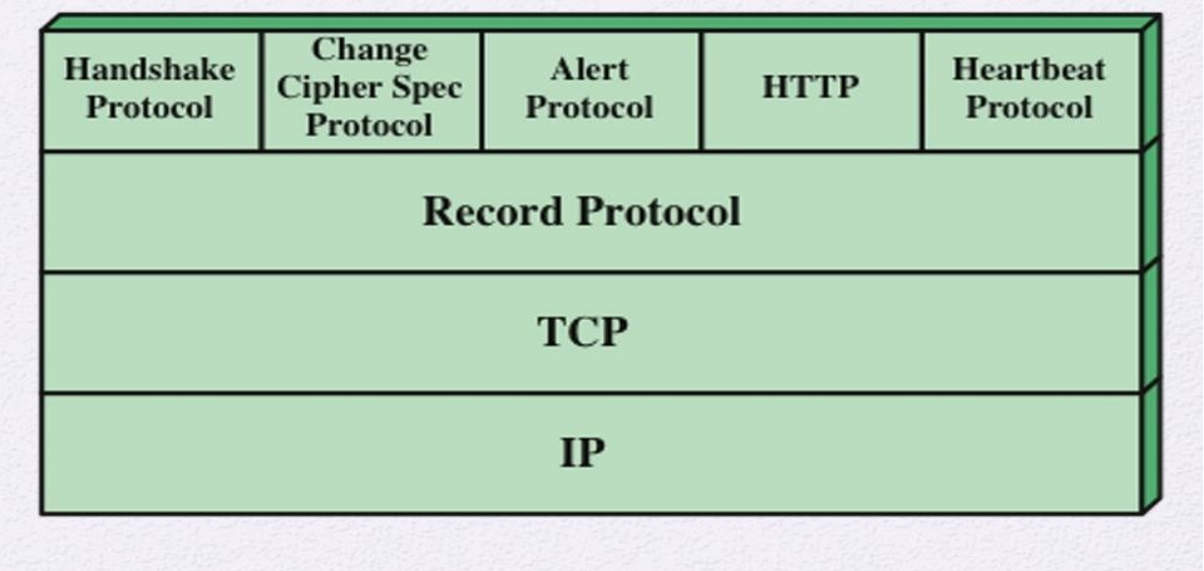


Figure 17.2 SSL/TLS Protocol Stack

TLS Record Protocol:

The Transport Layer Security (TLS) Record protocol secures application data using the keys created during the Handshake. The Record Protocol is responsible for securing application data and verifying its integrity and origin. It manages the following:

1. Dividing outgoing messages into manageable blocks, and reassembling incoming messages.
2. Compressing outgoing blocks and decompressingincoming blocks (optional).
3. Applying a Message AuthenticationCode(MAC) to outgoing messages, and verifying incoming messages using the MAC.
4. Encrypting outgoing messages and decrypting incoming messages.

SSL Record Protocol:

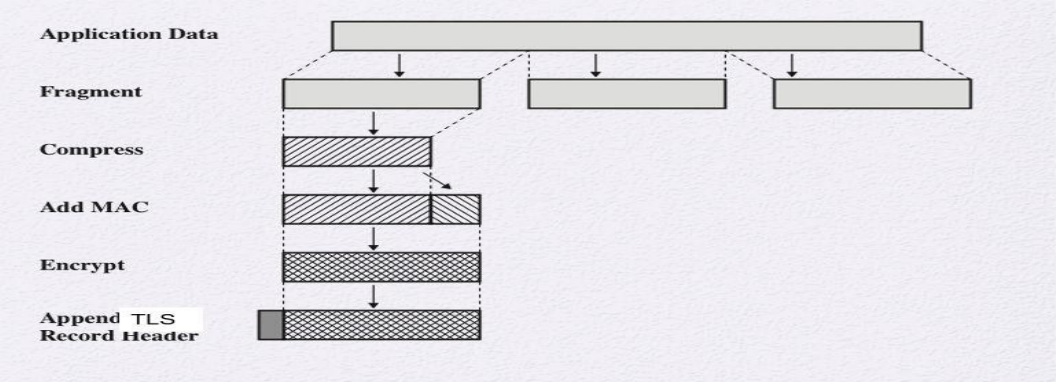


Figure 17.3 TLS Record Protocol Operation

SSL Record Protocol:

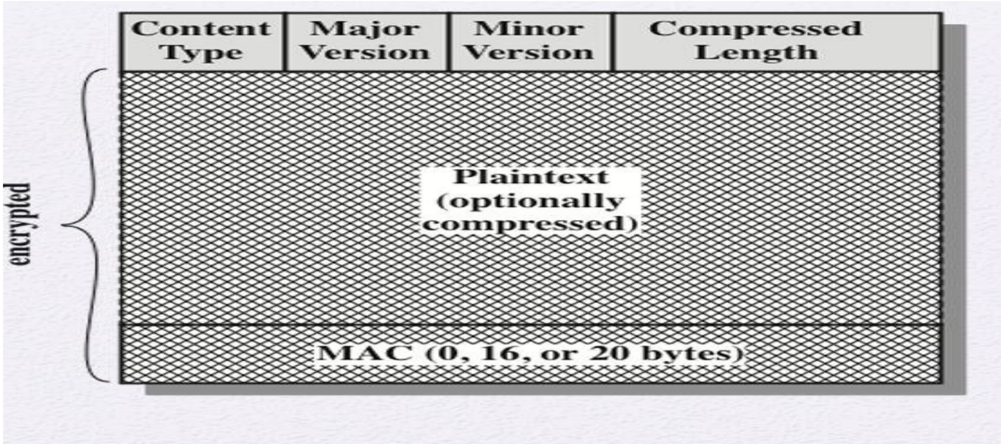
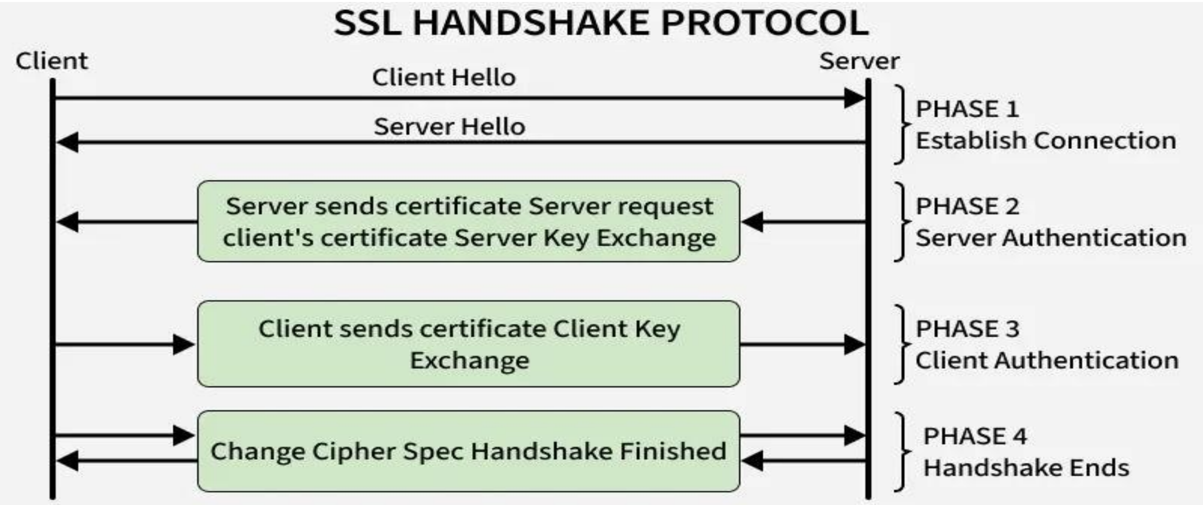


Figure 17.4 TLS Record Format

Handshake Protocol:

Establishes SSL sessions and authenticates clients and servers.



- Client and server exchange hello packets, protocol versions and cipher suites.
- Server sends its certificate and server key information.

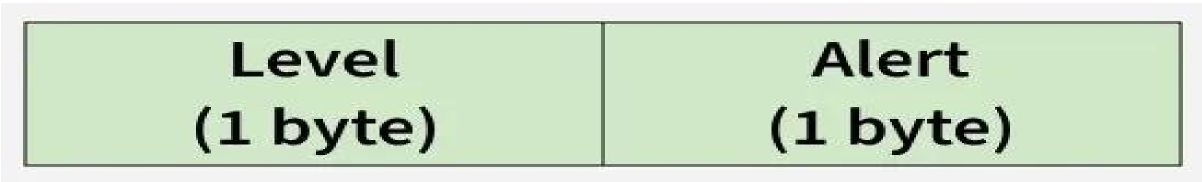
- Client responds with its certificate and key exchange.
- Change Cipher Spec finalizes the handshake, activating secure communication.

Change-Cipher Spec Protocol:

- Signalsthat pending cryptographic parameters from the handshake should now become active.
- Consists of a single 1-byte message.
- 

Alert Protocol:

- Communicates SSL-related warnings or errors.
- **Warning alerts** (level 1): Non-critical issues, such as expired or unsupported certificates.
- **Fatal alerts** (level 2): Critical errors, such as handshake failures, bad record MAC or illegal parameters, which terminate the connection.



Versions of SSL/TLS:

Version	Release Year	Notes
SSL 1	Never released	Insecure
SSL 2	1995	First public release
SSL 3	1996	Improved security
TLS 1.0	1999	Successor to SSL 3.0
TLS 1.1	2006	Improved encryption and security
TLS 1.2	2008	Widely adopted, strong encryption
TLS 1.3	2018	Modern, efficient, secure protocol

HTTP:

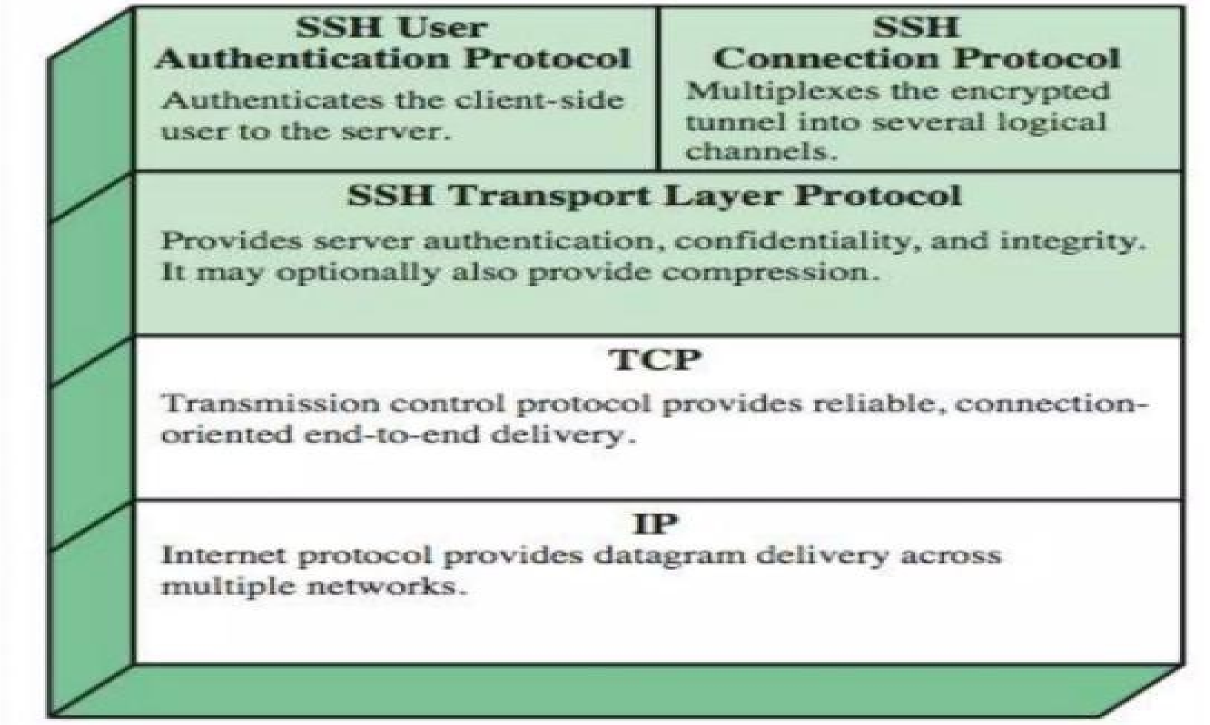
- HTTP acts as the application-level protocol that sits on top of the secure layer, forming HTTPS.
- It is the data being secured, while SSL/TLS handles encryption, authentication, and data integrity below it.
- HTTP initiates the request, and SSL/TLS securesthe data transfer**.**

HTTPS

- It is a Combination of HTTP&SSL/TLS to Secure Communications Between Browser and server
- Use https://URL rather than http://Url
- Connection Initiation
    - TLS handshake then http request(s)
- Connection Closure
    - Connection close in http

Secure Shell (SSH):

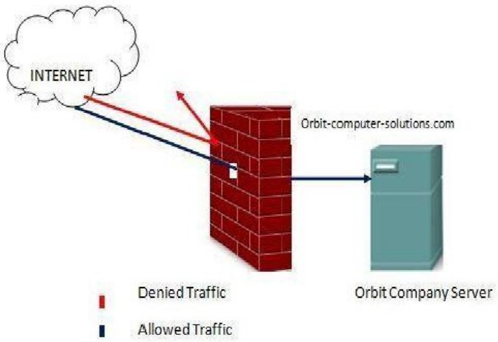
- SSH protocol for secure network communications between the client & server
- Designed to be simple & inexpensive
- SSH1 provided secure remote logon facility
- also has more general client/server capability
- SSH fixes a number of security flaws
- SSH clients & servers are widely available



Firewalls:

- Firewall is device that provides secure connectivity between networks or Client & Server.
- It is used to implement a security policy for communication between networks.
- A firewall may be a hardware, software that is used to prevent unauthorized access or internet users from accessing a private network or a single computer.
- All messages entering or leaving the intranet pass through the firewall, which examines each message & blocks those that do not meet the specified security criteria.

Firewall Characteristics and Access Policy



- All traffic from inside to outside, and vice versa, must pass through the firewall.
- Only authorized traffic, as definedby the local security policy, will be allowed to pass.
- The firewall itself is immune to penetration.
- Service Control: Determines the types of Internet services that can be accessed, inbound or outbound.
- Direction Control: Determines the direction in which particular service requests may be initiated and allowed to flow through the firewall.
- User Control: Controls access to a service according to which user is attempting to access it.

Why do we need a firewall?

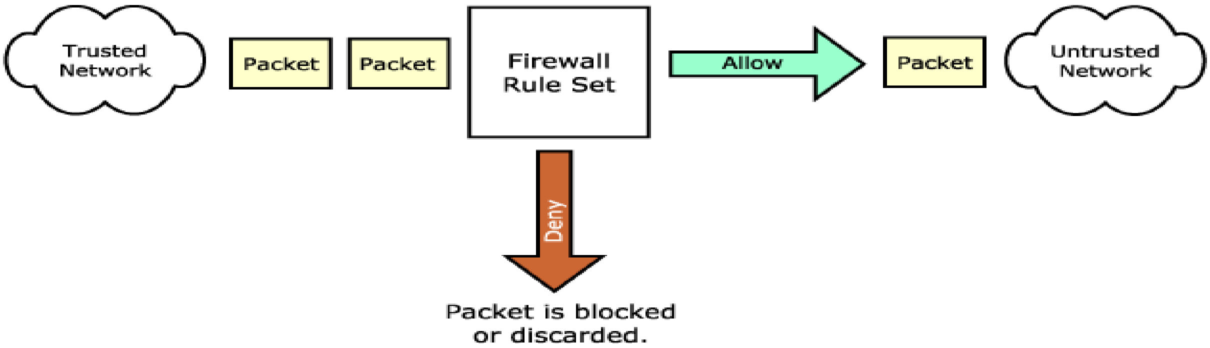
- To protect confidential information from those who do not explicitly need to access it.
- To protect our network & its resources from malicious users & accidents that originate outside of our network.

Types of firewall technique:

- Packet filter
- Application gateway
- Circuit-level gateway
- Bastion host

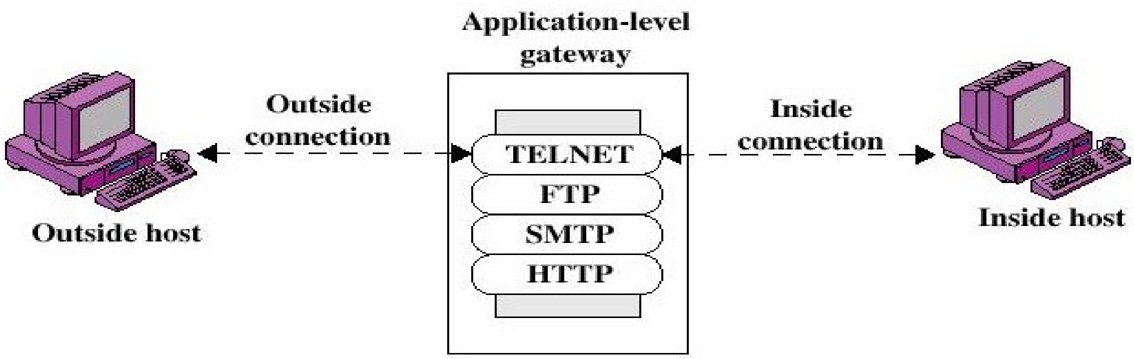
Packet filter:

- It looks at each packet entering or leaving the network and accepts or rejects it based on user-defined rules.
    - Source IP address
    - Destination IP address
    - Source & destination transport-level address(TCP or UDP)
- Packet filtering is fairly effective & transparent to users



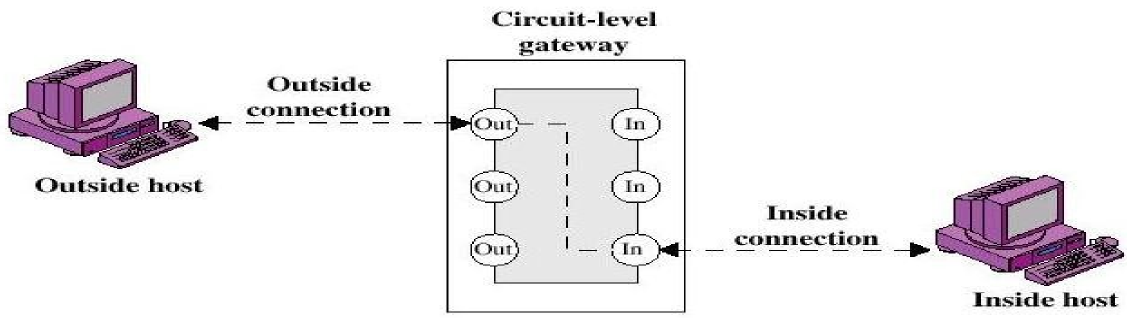
Application gateway:

- In such type of firewall remote host or network can interact only with proxy server, proxy server is responsible for hiding the details of the internal network i.e. intranet.
- Users uses TCP/IP application, such as FTP & Telnet servers.

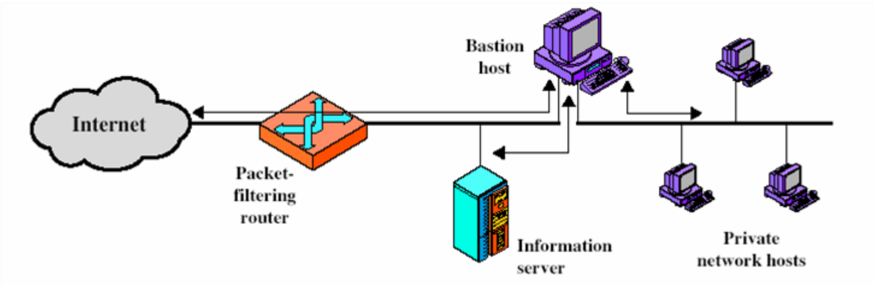


Circuit – levelFirewall:

- This can be a stand – alone system or it can be a specialized functions performed by an application – level gateway for certain applications.
- It does not permit an end – to – end TCP connection; rather, the gateway sets two TCP connections.
- A typical use of the circuit – level gateway is a situation in which the system administrator trusts the internal users.
- The gateway can be configured to support application level or proxy service on inbound connections and circuit level functions for outbound connections.



Bastion Host:

- Bastion host is a special purpose computer on a network specifically designed and configured to withstand attacks.
- It generally hosts a single application, provides platform for application gateway and circuit-level gateway.
- It supports limited/specific applications to reduce the threat to the computer.
- Include application-Telnet, SMTP, FTP
- 

(b) Screened host firewall system (dual-homed bastion host)

TYPES OF FIREWALLS:

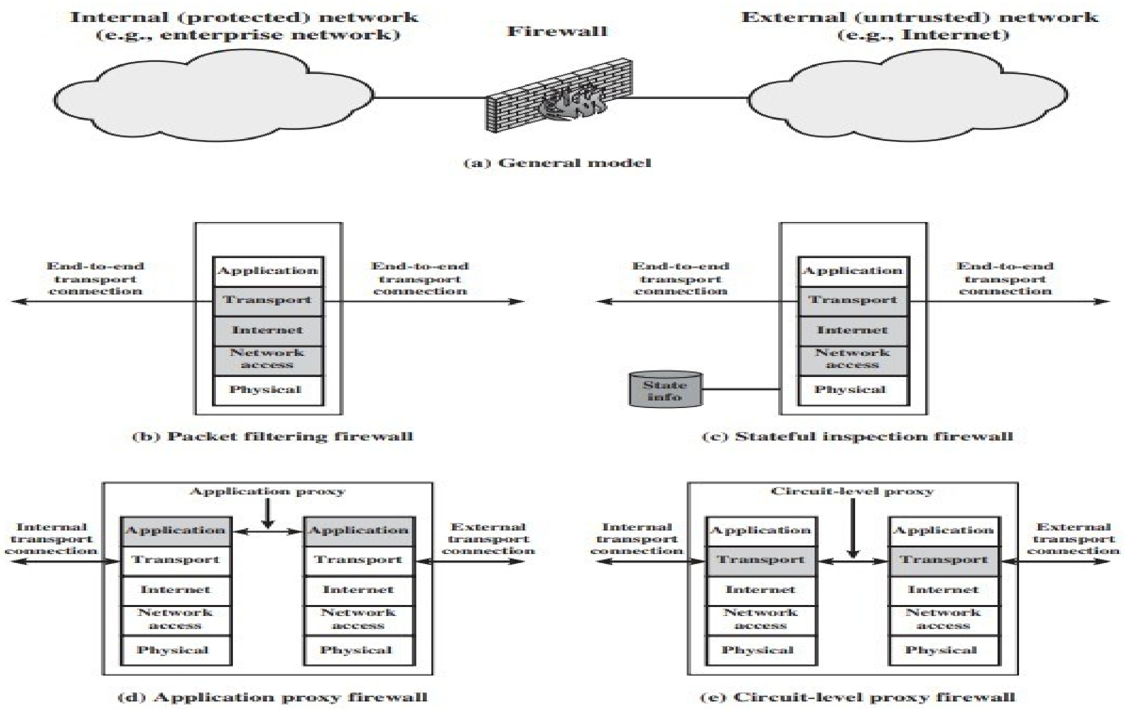


Figure 22.1 Types of Firewalls

FIREWALL SETTING: